

Smooth and Nonsmooth Markowitz Portfolio Optimization

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I. INTRODUCTION

Portfolio optimization is the core of modern quantitative finance. It helps investor solve the trade-off problem between maximizing expected returns from investments while managing risks. Originally, this problem was described as a smooth mean-variance problem introduced by Markowitz [1]. This problem minimizes the portfolio variance while targeting a certain expected return. This first problem, however, does not take *transaction cost* into account, therefore leading to a second version of this problem integrating this in the objective. This new problem is more realistic, though it is now non-smooth. Because of this key difference between the two models, the suitable optimization methods will differ from one another.

The aim of this report is thus to compare different methods for both models by analyzing the cost of the methods, their convergence—both empirically and using theoretical results—and their performance on historical data from the S&P500.

II. DATA

How did you handle the raw data?

To manage the data, we imported the CSV file via Panda in order to convert it into a data frame. We then transformed the dataset so that each column corresponds to the closing price of stock X and the corresponding row corresponds to the date.

How did you estimate μ and Σ ?

To estimate stock returns and the covariance matrix, we used the following formula : $r_t = \log\left(\frac{C_t}{C_{t-1}}\right)$ where r_t correspond to the return at time t and C_t correspond to the close time at t .

A positive semi-definite covariance matrix would greatly improve the model. Why? Is it PSD? If not, can you make it naturally?

III. SMOOTH MODEL

A. Model

B. Projected Gradient Descent

Description of the methods

What do you need for each of them?

Compare the theory with some first numerical results.

What can be improved compared to the theory? Why? Is it normal?

For example:

Input: step size $\alpha > 0$
Output: approximate solution x_N

for $k = 0, 1, \dots, N - 1$ **do**
 compute a gradient g_k
 $x_{k+1} = x_k - \alpha g_k$
end for

Algorithm 1. Gradient Descent

I need a gradient and a step size for Algorithm 1 to work, so the gradient is ... and a classical choice of step size is ..., so I need to compute

C. Projected Gradient Descent with Momentum

What is the momentum doing?

D. Projected Randomized Coordinate Descent

Is it smart to make deterministic choices for the coordinates? Is the answer the same in theory and in practice? Discuss it.

IV. NON-SMOOTH MODEL

A. Model

Key differences with the smooth case. What fundamental changes are you expecting? Do you verify them?

B. Projected Subgradient Method

C. Proximal Gradient Descent

Did you think of other methods? Why could they help solve the problem? What structure of the problem made you think of this method?

V. NUMERICAL RESULTS

What comparisons could you make between the different methods? Is it always a fair choice?

Can some methods be greatly improved compared to the theory?

Are some of them disappointing? Do you have an explanation?

In general, does a model (smooth or non-smooth) bring better solutions? What do you mean by better? Are the methods faster? Do the solutions have a particular structure? Is it normal?

VI. CONCLUSION

Summary.

REFERENCES

- [1] H. Markowitz, "Portfolio Selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.